Timothy Do's Official CV

An A.I. Machine Vision Enthusiast & DSP Techie to Make the World A Better Place

📞+1-(408) 644-5538 📩timothydobsa@gmail.com 🤝Linkedin (/in/do-timothy) 💻Github (dotimothy)

EDUCATION

University of California Irvine, Irvine, CA — *B.S. Electrical Engineering*

September 2019 - June 2023 (Expected)

- 4th Year Undergraduate, Accelerated Status
- Overall GPA: 3.90, Major GPA: 3.91
- **Specialization**: Digital Signal Processing
- **Research Interests:** Digital Signal Processing with Artificial Intelligence, Digital Image Processing, Machine Vision
- **Relevant Coursework:**
    - EECS 22: Advanced C Programming, learned about advanced data structures, program decomposition (Winter 2021)
    - EECS 22L: Introduction to Software Engineering, worked in a team of 4 (TTLZ22s) to develop a network chess application in C with a git repo with version control (Spring 2021)
    - EECS 50/150: Discrete-Time/Continuous Signals and Systems, learned about LTI Systems, Z-Transforms, the DTFT, FFT, Anti-Aliasing, Laplace Transform (Spring 2021/Winter 2022)
    - EECS 152A/B: Digital Signal Processing, learned about the Fourier Transform, Sampling, Filter Design (Fall 2021)
    - EECS 203A: Digital Image Processing, learned about Image transforms, 2D Fourier Transform, MATLAB processing. (Spring 2022)
    - EECS 170A/B/C: Electronics, learned about semiconductor properties and the topologies of diodes, BJT, and MOSFET transistors and their applications
- **Major Coursework** for Fall 2022: EECS 159A (Senior Design Project), EECS 160A/LA (Control Systems), ENGR 190W (Engineering Writing)
- **Awards:** Dean's Honor List (GPA 3.5+) for 9 Quarters (Fall 2019, Winter 2020, Spring 2020, Fall 2020, Winter 2021, Spring 2021, Fall 2021, Winter 2022, Spring 2022)

Evergreen Valley High School, San Jose, CA — *Diploma*

August 2015 - May 2019

- GPA: Unweighted: 3.8, Weighted: 4.0
- **Relevant Coursework**: AP Physics C: Mechanics (4), AP Computer Science A (4), AP Statistics (4), AP Calculus AB (5), AP US History (4), Math 072 (Calc II) @ Evergreen Valley College
- **Awards:** EVHS Band Scholar, Magna Cum Laude, Golden State Diploma, Bilingual Certification in Vietnamese, AP Scholar with Distinction

SKILLS

Musicianship:
Played the Euphonium in an ensemble setting

Teaching:
Created teaching plans for music tutoring & lead sections of the band during rehearsal

Planning:
Planned and delegated the blueprint design cabinets for my Eagle Scout Project

Web Programming:
Created Websites with HTML, CSS, and Javascript

Compiler / Object-Oriented Programming:
Experience with C, Java, and Python libraries

GPIO Programming:
Used MicroPython or Ino to control circuits digitally with Arduino or Raspberry Pi.

Photo/Video Editing:
Edited Videos/Photos as Final Cut Pro, Photoshop, & Premiere Pro.

Computers:
Assembled/Debugged Desktop machines with OS imaging.

CUDA:
Programmed GPUs to do Matrix Multiplications and DSP

ACTIVITIES

Vice President of IPC @ UCI

Member of UCI IEEE HKN, TBP

Student Researcher for ICVL (Irvine Computer Vision Laboratory)

Former 3rd Chair Euphonium Player for the 2019 Santa Clara County Honor Band

Former Principal Euphonist for the 2019 EVHS Wind Ensemble

EXPERIENCE

Most Holy Trinity Parish, San Jose, CA — *Teacher's Assistant*

September 2013 - May 2019

- Helped young students in their religious studies in Catholicism and their skills in Vietnamese.

Boy Scouts of America Troop 610, San Jose, CA — *Eagle Scout*

September 2013 - September 2019

- Awarded the Bronze Palm: Earning 5+ additional Merit Badges over the 21 required for Eagle Scout.
- **Eagle Scout Project**: Constructing 2 Food Cabinets for the Milpitas Food Bank

Evergreen Music Mentors, San Jose, CA — *Founding President*

September 2018 - September 2021

- Lead music tutoring sessions for younger middle school students
- Integrated software (Cybernotes) as a video calling platform

Environmental Injustice , Irvine, CA — *Collaborator*

June 2020 - September 2020

- Developed techniques such as stakeholder analysis, using Civic Data Tools such as CalEnviroScreen , and citing in Chicago format.

Boy Scouts of America, San Jose, CA - *Merit Badge Counselor*

July 2021 - Present

- Offered to teach five Merit Badge Courses: Programming, Music, and Bugling, Electronics, and Digital Technology.

UC Irvine D.C.E., Irvine, CA - *I.T. General Assistant*

July 2021 - June 2022

- Helped Set-Up Desktops, Docks, Webcams, and Computers for over 300+ Employees in UCI's Division of Continuing Education

Western Digital, San Jose, CA - *Industrial IOT Intern*

June 2022 - September 2022

- **Skills:** MySQL, Python, Raspberry Pi, Soldering, Linux, Lean Six Sigma
- Certified as a Six Sigma Yellow Belt for learning Six Sigma foundations.
- Developed Presentation Skills in a Corporate Environment.
- Learned Industrial Tool Sets for Circuit Design and Embedded System Programming.

UC Irvine, Irvine, CA - Student Researcher

January 2022 - Present

- Assisted in research under the direction of Professor Glenn Healey on the topic of Spectral Image Filtering for Emissions on Wildfires
- Digitized Imaging Filters into CSVs and compared them in MATLAB against certain chemical emissions.
- Generated altitude maps using QGIS & SRTM 1 Arc-Second Global

IPC Club @ UCI, Irvine, CA - *Vice President*

November 2021 - Present

- Reserved club meeting rooms and sent weekly emails to club members regarding club events
- Programmed the club website and managed social media (i.e. moderator of club discord server)
- Led a Two-Part AM Radio Workshop where I had to design the circuit for the best amplifier gain in addition to creating easy-to-read slides & ordering parts from Amazon/Dollar Tree.

PUBLICATIONS

Timothy Do; Matthew Prata; Jorge Radge; Alex Wang. 2022. Colorization Utilizing Unsupervised Methods.UCI EECS 195, Network Science. UCI Henry Samueli School of Engineering. University of California, Irvine.

Tahis Alcantar; **Timothy Do**; April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. Fast Disaster Case Study: Santa Fe Springs. UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California, Irvine.

Tahis Alcantar; **Timothy Do;** April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. Slow Disaster Case Study: Wilmington. UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California, Irvine.

Tahis Alcantar; **Timothy Do**; April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. Combo Disaster Case Study: San Bernardino County. UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California Irvine

LANGUAGES

English (Fluent), **Vietnamese** (Fluent), & **Spanish** (Proficient)

HACKATHONS

Cisco Webex Hackathon 2021

Gratition: An Easing Mental Health Web App (with Google Voice Synthesis)
Languages: HTML/CSS/JS

ZotHacks 2021

Anteatings: A Fun JS-Based Antsmashing Game on the Web
Languages: HTML/CSS/JS

PROJECTS

TTLZNetChess - *An Advanced Client-Based Chess Application*

March 2021 - June 2021

- Managed a team of four developers into develop a C-Based Chess Application for EECS22L (Intro to Software Engineering)
- Implemented Terminal Graphics (Colors) using Escape Codes
- Implemented AIs in Chess using Minimax Algorithm
- Networking Support using standard sockets and FCFS Protocol

RefreshVideos — *A High Refresh Rate Video Player*

April 2020 - Present

- **Languages Used**: HTML / CSS / Javascript
- Used Canvas to play videos at high frame rates and apply effects
- Featuring filters like Edge Detection, Aging, etc.

pdDSP — *Audio-Processing using Pure-Data*

March 2020

- **Languages Used**: Pure Data
- Used Pure Data to mix audio and designed audio filters (e.g. Low Pass Filter, Band Pass Filter)

PiFmMorse — *A Morse Code FM Transmitter for the Raspberry Pi*

September 2020

- **Languages Used**: Python
- An Program to transmit Morse Code over FM radio via GPIO 4
- Learned to concatenate .wav files via wave library
- Learned to call linux commands using os library

RPiMorse — *A Morse Code LED Circuit for the Raspberry Pi*

September 2020

- **Languages Used**: Python
- An Program to flash an LED in Morse Code given a circuit
- Learned GPIO Pins Wiring and its respective python library

Timduino Tutorials — *Experiments with the Arduino Uno*

September 2020

- **Languages Used**: C
- Learned functions to set voltages (pinMode(), digitalWrite())
- Wired Circuits and pins to apply Arduino Programs

SCU Complex Fires — *A Website Providing Wildfire Resources*

August 2020 - September 2020

- **Languages Used**: HTML / CSS/ Javascript
- Provided Wildfire resources for research, shelter, and news
- Worked with a team of 4 and remotely collaborated through Github
- Features an Energy-Efficient Dark Mode.